1. **Project Title - Student Performance Analytics System**

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   Technology - Python, NumPy, Pandas

2. **Objective**

   The objective of this project is to develop a Python-based Student Performance Analytics System that analyzes student marks and attendance using Pandas and NumPy. The system calculates total and average marks, assigns grades, identifies pass/fail students, performs subject-wise analysis, and identifies top-performing students

3. **Technology used**
   - Python - Used for implementing the main program and logic
   - NumPy - Used for numerical calculations and data processing
   - Pandas - Used for loading, organizing, and analyzing the student dataset
   - Git - Used for version control
   - GitHub - Used to store and share the project repository

4. **Dataset Description**

   The project uses a CSV dataset named students.csv containing 20 student records. The dataset contains student information, department, marks for different subjects, and attendance percentage. The subject marks are used to calculate total marks, average marks, grades, and pass or fail results
   The subject columns are:
   - Student_ID
   - Name
   - Department
   - Python
   - Mathematics
   - Data analysis

- Computer fundamentals
- Communication
- Attendance

5. Implementation

- **Loading the Dataset -** The student dataset is loaded from the students.csv file using the Pandas read_csv() function
- **Organizing the Data** - The data is stored in a Pandas DataFrame, which makes it easier to access, process, and analyze student records
- **Calculating Total Marks** - The marks of all subjects are added for each student using a reusable function called calculate_total()
- **Calculating Average Marks -** The average marks of each student are calculated using the calculate_average() function
- **Assigning Grades -** Grades are assigned based on the student's average marks using the assign_grade() function
- **Checking Pass/Fail -** The check_result() function checks the marks of each subject. A student is considered passed if they score at least 40 marks in every subject
- **Overall Class Analysis -** The system calculates the overall class average and identifies the highest and lowest performing students
- **Subject-wise Analysis** - The average marks of each subject are calculated to identify the best and lowest performing subjects
- **Top-performing Students -** The system sorts students according to their average marks and displays the top five performing students
- **Attendance Analysis -** The system calculates average attendance and identifies students whose attendance is below 75%

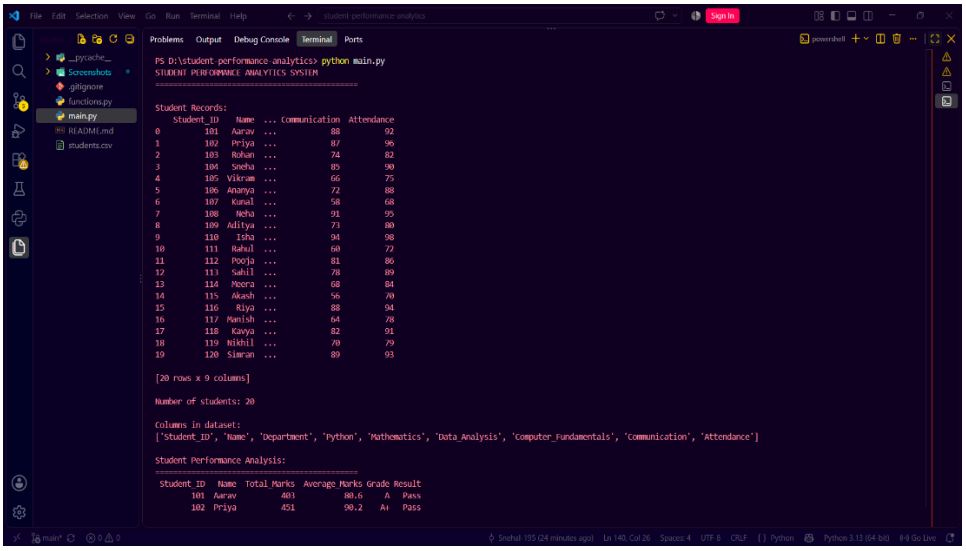
6. Key Features

- Student dataset loading using Pandas
- Total marks calculation
- Average marks calculation
- Grade assignment
- Pass/fail identification
- Class average calculation
- Highest and lowest performer identification
- Subject-wise performance analysis
- Top 5 performing students

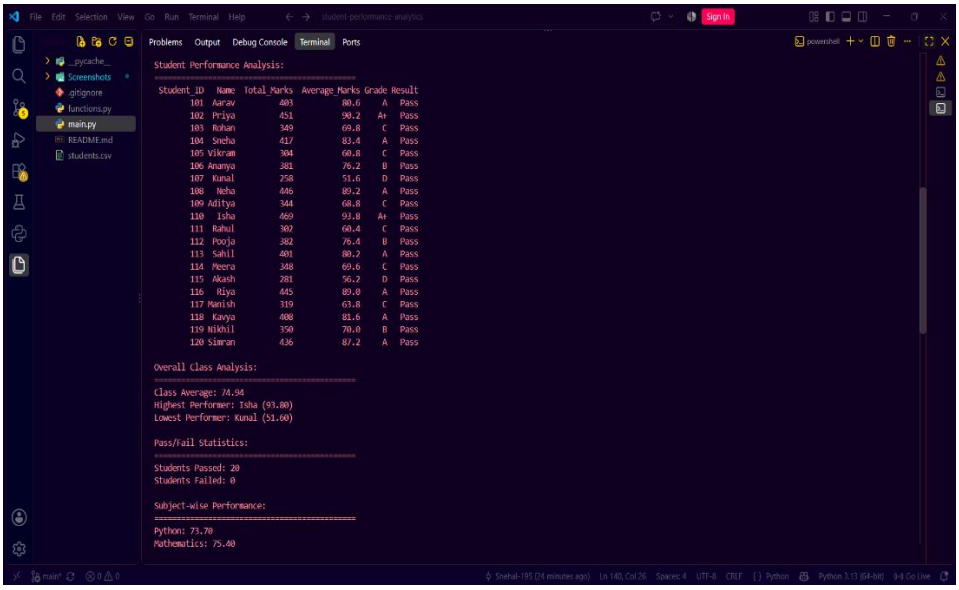
- Attendance analysis
- Use of reusable Python functions
- Use of loops and conditional statements

7. Output / Screenshots

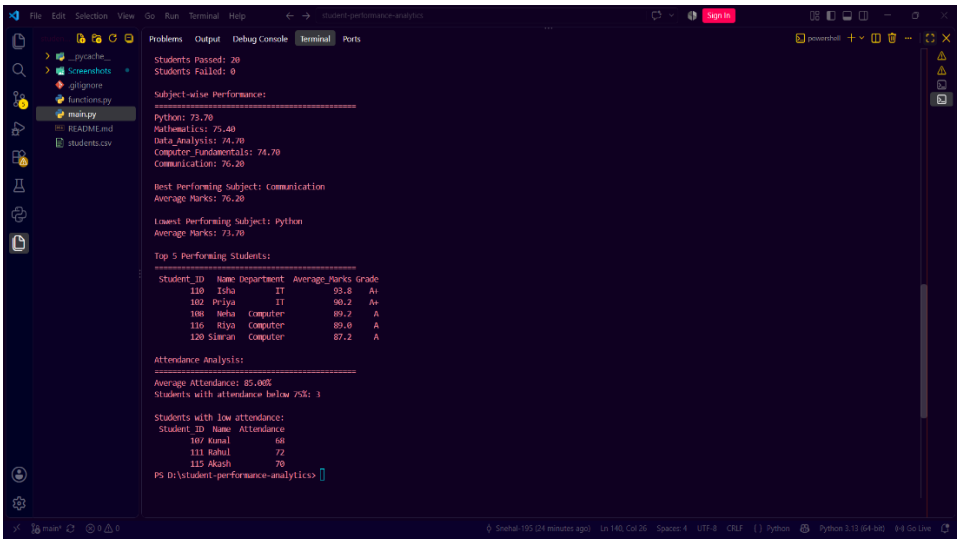
- **Figure 1: Student Records and Datasheet Information**



- **Figure 2: Student Performance Analysis and Class Analysis**



- **Figure 3: Subject-wise Performance, Top Students and Attendance Analysis**



8. Final Outcome

The Student Performance Analytics System successfully processes student data and provides meaningful performance insights. The system calculates individual student performance, class-level statistics, subject-wise averages, grades, pass/fail results, top-performing students, and attendance information. The project demonstrates the practical use of Python, NumPy, Pandas, functions, loops, and conditional statements for basic data analysis

9. Challenges and Learning

During this project, I learned how to work with CSV datasets and Pandas DataFrames. I also improved my understanding of Python functions, loops, conditional statements, and NumPy calculations. The project helped me understand how programming concepts can be applied to perform practical data analysis. I also learned how to organize a project and upload it to GitHub

10. GitHub Repository

https://github.com/Luffy093/student-performance-analytics